

Data Card: Gemma-2B Diabetes Dataset

AI Saturdays Lagos Cohort 10| Fine-Tuning Corpus Specifications

1. DATASET OVERVIEW & ORIGINS

The training dataset is a consolidated, curated corpus of **6,192 medical instruction and Q&A instances** specifically focused on diabetes prevention, diagnostic criteria, clinical management, and lifestyle modifications. The data was aggregated from three distinct sources to balance clinical accuracy and natural conversation.

Dataset File	Sample Count	Focus Area	Primary Content Type
diabetes_QA_dataset.csv	~692 pairs	Clinical Q&A	Direct diagnostic and clinical management questions
diabetes_instruct_temp_v44.csv	~500 pairs	Instruction Following	Contextual patient queries and step-by-step guidance
ada_diabetes_5000_instruction.csv	~5,000 pairs	ADA Guidelines	American Diabetes Association standards & advice

2. DATA STRUCTURING & PROMPT FORMATTING

All raw CSV pairs were unified into Gemma's turn-based chat template to ensure strict instruction-following behavior during fine-tuning:

```
<start_of_turn>user You are a domain-specific assistant specialized in diabetes care, guidance, and medical context. Provide accurate, helpful, and concise responses. User Query: [Instruction / Context]<end_of_turn> <start_of_turn>model [Target Ground Truth Response]<end_of_turn>
```

3. QUALITY CONTROL & PREPROCESSING

Deduplication & Cleaning

Removed redundant entries across the three files, stripped empty values, and standardized whitespace and medical terminology encoding.

Length Optimization

Truncated entries exceeding maximum context lengths to avoid truncation artifacts while preserving core clinical guidance.

4. LIMITATIONS & ETHICAL CONSIDERATIONS

- Geographic Bias:** Predominantly based on Western clinical standards (ADA guidelines); may require adaptation for local dietary contexts.
- Synthetic/Aggregated Nature:** Synthetic instruction templates require validation against real-world patient query